CONSULTANT TECHNICAL DUE DILIGENCE REPORT

mdadeel/Arbor

Project: Arbor

PREPARED FOR **Northstar Technical Advisory**

REPOSITORY
mdadeel/Arbor

BRANCH
arena/d8319581-arbor

PINNED COMMIT SHA
b4d9090aec645d37c546226d656d0afcc027b146

SCAN DATE
2026-10-28 14:01:34 UTC

SNAPSHOT SHARE EXPIRY
2026-11-04 14:01:34 UTC

EFFECTIVE POLICY VERSION
security@1.0.0, maintainability@1.0.0, accessibility@1.0.0

C

Technical Debt Warning

Composite Score: **64** / 100

5 critical · **23** warnings · **5** info

DIMENSION	MEASURED SIGNAL SCOPE	SCORE
Architecture	Module boundaries, file density, and cyclic import discipline	95/100
Maintainability & Debt	Type safety, dead exports, unused dependencies, and debug hygiene	14/100
Performance Signals	Server/client component balance and media/asset patterns	94/100
Documentation & Env	README completeness, inline documentation, and .env templates	78/100
Security & Supply Chain	Secret pattern checks and known dependency advisory exposure	15/100
Design System	Semantic token adoption, variant consistency, and color hygiene	59/100

Methodology & Snapshot Integrity: Arbor evaluates source ASTs, module import graphs, lockfiles, and deterministic policy rules statically without executing repository code. Scores and findings apply strictly to commit b4d9090aec64 scanned on 2026-10-28; verify against current HEAD before release sign-off.

SECTION 01

Recommended Actions

Effort: S (<2h) · M (0.5–2d) · L (3d+)

Top 5 highest-leverage remediations derived deterministically from finding severity, module import fan-in, and repository blast radius.

#	FINDING & BLAST RADIUS	RECOMMENDED REMEDIATION	EFFORT
1	CRITICAL **Possible Hardcoded credential committed** src/lib/auth.ts:125 High fan-in · 15 dependents	Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.	L
2	CRITICAL **Possible Generic secret assignment committed** src/lib/auth.test.ts:6 Test suite scope	Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.	S
3	CRITICAL **Possible Hardcoded credential committed** src/lib/auth.test.ts:6 Test suite scope	Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.	S
4	CRITICAL **Possible Hardcoded credential committed** src/lib/github-webhook.test.ts:6 Test suite scope	Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.	S
5	CRITICAL **Possible Hardcoded credential committed** src/server/services/ai.test.ts:38 Test suite scope	Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.	S

SECTION 02

Architecture Summary & Codebase Posture

Full Budget Coverage · 277/277 files parsed

ARCHITECTURAL STRENGTHS

- Centralized design token system detected (tailwind-theme).
- Modular source layout spanning 277 source files and 40,123 lines of code.

FOCUS & RISK AREAS

- 5 critical security/credential signal(s) require immediate triage.
- 1 circular dependency chain(s) coupling module boundaries.
- 230 explicit any annotations weakening static type guarantees.
- 12 potentially unused package(s) declared in manifest.
- 20 exported symbol(s) without internal static importers.
- 142 hardcoded color literal(s) bypassing semantic tokens.

DETECTED STACK:

Next.js · npm · TypeScript · JavaScript · Prisma · Redis · Vitest · Playwright · Testing Library · Tailwind CSS · shadcn/ui · Radix UI · Lucide Icons

SCALE & DENSITY		COMPONENT ARCHITECTURE		HYGIENE & DEBT		GRAPH & DESIGN SYSTEM	
Source Files	**277**	Total Components	**127**	Explicit any	**230**	Graph Modules	**277**
Lines of Code	**40,123**	Server Components	**66**	Console Logs	**15**	Import Edges	**755**
Avg File Lines	**145**	Client Components	**61**	Unused Deps	**12**	Circular Loops	**1**
Indexed Size	**2.9 MiB**	Custom Hooks	**216**	Dead Exports	**20**	Hardcoded Colors	**142**

Directory File Distribution

DIRECTORY	FILES
src/server/services	51
src/lib	30
src/components/dashboard	24
src/server/analysis	24
src/server/routers	18
src/components/ui	16

Largest Source Modules (>300 LOC)

MODULE PATH	LINES
src/components/dashboard/admin/admin-view.tsx	1,546
src/server/services/admin.ts	1,210
src/components/dashboard/project-report.tsx	1,178
src/components/marketing/product-screens.tsx	796
src/components/dashboard/api-explorer.tsx	762
src/components/dashboard/printable-project-report.tsx	626

SECTION 03

Detailed Findings by Severity

33 active (5C / 23W / 5I)

CRITICAL Critical Severity Findings (5)

High-impact security or structural risks requiring immediate remediation

Possible Hardcoded credential committed

security | security@1.0.0 | src/lib/auth.test.ts:6

A value matching this pattern was found in source text. The scanner cannot tell whether it is live, revoked, or a test fixture.

Evidence: src/lib/auth.test.ts:6

Remediation: Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.

Possible Generic secret assignment committed

security | security@1.0.0 | src/lib/auth.test.ts:6

A value matching this pattern was found in source text. The scanner cannot tell whether it is live, revoked, or a test fixture.

Evidence: src/lib/auth.test.ts:6

Remediation: Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.

Possible Hardcoded credential committed

security | security@1.0.0 | src/lib/auth.ts:125

A value matching this pattern was found in source text. The scanner cannot tell whether it is live, revoked, or a test fixture.

Evidence: src/lib/auth.ts:125

Remediation: Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.

Possible Hardcoded credential committed

security | security@1.0.0 | src/lib/github-webhook.test.ts:6

A value matching this pattern was found in source text. The scanner cannot tell whether it is live, revoked, or a test fixture.

Evidence: src/lib/github-webhook.test.ts:6

Remediation: Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.

Possible Hardcoded credential committed

security | security@1.0.0 | src/server/services/ai.test.ts:38

A value matching this pattern was found in source text. The scanner cannot tell whether it is live, revoked, or a test fixture.

Evidence: src/server/services/ai.test.ts:38

Remediation: Verify the value without copying it into tickets or logs. If real, revoke/rotate it, remove it from active code, and audit repository history and access.

WARNING Warning Severity Findings (23)

Maintainability, accessibility, boundary, and configuration hazards

Button may not have an accessible name

accessibility | accessibility@1.0.0

src/components/dashboard/admin/admin-navbar.tsx:122

src/components/dashboard/admin/admin-navbar.tsx:122 contains an icon-only or empty button without a detectable accessible-name attribute.

Evidence: src/components/dashboard/admin/admin-navbar.tsx:122

Remediation: Provide visible text or an aria-label/aria-labelledby that describes the action, then verify the rendered accessibility tree.

Button may not have an accessible name

accessibility | accessibility@1.0.0 | src/components/ui/createui/button.tsx:162

src/components/ui/createui/button.tsx:162 contains an icon-only or empty button without a detectable accessible-name attribute.

Evidence: src/components/ui/createui/button.tsx:162

Remediation: Provide visible text or an aria-label/aria-labelledby that describes the action, then verify the rendered accessibility tree.

Image may be missing alternative text

accessibility | accessibility@1.0.0 | src/server/analysis/accessibility.test.ts:6

src/server/analysis/accessibility.test.ts:6 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/accessibility.test.ts:6

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Button may not have an accessible name accessibility accessibility@1.0.0 src/server/analysis/accessibility.test.ts:22

src/server/analysis/accessibility.test.ts:22 contains an icon-only or empty button without a detectable accessible-name attribute.

Evidence: src/server/analysis/accessibility.test.ts:22

Remediation: Provide visible text or an aria-label/aria-labelledby that describes the action, then verify the rendered accessibility tree.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/ast.ts:115

src/server/analysis/ast.ts:115 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/ast.ts:115

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/checks.ts:198

src/server/analysis/checks.ts:198 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/checks.ts:198

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/checks.ts:208

src/server/analysis/checks.ts:208 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/checks.ts:208

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/checks.ts:328

src/server/analysis/checks.ts:328 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/checks.ts:328

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/checks.ts:329

src/server/analysis/checks.ts:329 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/checks.ts:329

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/checks.ts:338

src/server/analysis/checks.ts:338 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/checks.ts:338

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/checks.ts:339

src/server/analysis/checks.ts:339 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/checks.ts:339

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Image may be missing alternative text accessibility accessibility@1.0.0 src/server/analysis/index.ts:224

src/server/analysis/index.ts:224 contains an <img> element without an alt attribute.

Evidence: src/server/analysis/index.ts:224

Remediation: Add concise, context-appropriate alt text, or explicitly use alt="" for a decorative image.

Large source file techDebt maintainability@1.0.0 src/components/dashboard/admin/admin-view.tsx

src/components/dashboard/admin/admin-view.tsx has 1546 non-empty lines. This is a maintainability hint, not a runtime-performance or correctness defect.

Evidence: src/components/dashboard/admin/admin-view.tsx

Remediation: Start with the exact evidence below; make small changes and validate with the project test suite.

Large source file techDebt maintainability@1.0.0 src/server/services/admin.ts

src/server/services/admin.ts has 1210 non-empty lines. This is a maintainability hint, not a runtime-performance or correctness defect.

Evidence: src/server/services/admin.ts

Remediation: Start with the exact evidence below; make small changes and validate with the project test suite.

Large source file techDebt maintainability@1.0.0 src/components/dashboard/project-report.tsx

src/components/dashboard/project-report.tsx has 1178 non-empty lines. This is a maintainability hint, not a runtime-performance or correctness defect.

Evidence: src/components/dashboard/project-report.tsx

Remediation: Start with the exact evidence below; make small changes and validate with the project test suite.

Large source file techDebt maintainability@1.0.0 src/components/marketing/product-screens.tsx

src/components/marketing/product-screens.tsx has 796 non-empty lines. This is a maintainability hint, not a runtime-performance or correctness defect.

Evidence: src/components/marketing/product-screens.tsx

Remediation: Start with the exact evidence below; make small changes and validate with the project test suite.

Large source file techDebt maintainability@1.0.0 src/components/dashboard/api-explorer.tsx

src/components/dashboard/api-explorer.tsx has 762 non-empty lines. This is a maintainability hint, not a runtime-performance or correctness defect.

Evidence: src/components/dashboard/api-explorer.tsx

Remediation: Start with the exact evidence below; make small changes and validate with the project test suite.

Circular dependency techDebt maintainability@1.0.0 Repository-wide

Import cycle: src/components/dashboard/finding-item.tsx → src/components/dashboard/finding-workflow-actions.tsx → src/components/dashboard/finding-item.tsx. This is a coupling signal, not proof of a runtime bug.

Remediation: Inspect the listed path and consider extracting shared types or utilities if the cycle is causing real coupling.

230 explicit `any` type usages techDebt maintainability@1.0.0 Repository-wide

Explicit any types weaken compile-time guarantees in the locations where they occur.

Remediation: Start with the exact evidence below; make small changes and validate with the project test suite.

Heavy client-component usage performance maintainability@1.0.0 Repository-wide

51% of React files are "use client"; prefer server components where possible.

Remediation: Use browser/build measurements to identify expensive client boundaries; keep interactive state client-side and move static work server-side.

Both next/image and raw <img> tags performance maintainability@1.0.0 Repository-wide

3 raw <img> tags found though next/image is available; they skip optimization.

Remediation: Measure the affected route/component first, then optimize only where a real user-facing cost is observed.

142 hardcoded color(s) in component files designSystem maintainability@1.0.0 Repository-wide

Prefer theme tokens over raw colors so the design system stays consistent.

Remediation: Adopt shared tokens and component conventions where they help the product remain consistent.

2 environment variable(s) are undocumented environment maintainability@1.0.0 Repository-wide

NODE_ENV, VERCEL

Remediation: Document variable names and safe example values; never add production secrets to the example file.

INFO Informational Signals (5)

Low-urgency hygiene observations and cleanup opportunities

12 possibly unused dependency(ies) techDebt maintainability@1.0.0 Repository-wide

@hookform/resolvers, @tanstack/react-table, @vitejs/plugin-react, autoprefixer, bcryptjs, eslint-config-next, jsdom, postcss, postgresql-client, react-dom

Remediation: Search the full repository and build configuration before removing it; then run tests and production build.

20 export(s) not found in static imports techDebt maintainability@1.0.0 Repository-wide

src/middleware.ts#middleware, src/middleware.ts#config, src/app/layout.tsx#metadata, src/app/not-found.tsx#metadata, src/app/opengraph-image.tsx#runtime, src/app/opengraph-image.tsx#alt

Remediation: Check package entry points and framework conventions before removing the export.

15 console.log calls techDebt maintainability@1.0.0 Repository-wide

Static scan found console.log calls; some may be intentional diagnostics.

Remediation: Start with the exact evidence below; make small changes and validate with the project test suite.

Very few comments documentation maintainability@1.0.0 Repository-wide

1% of lines are comments.

Remediation: Document the project setup, important workflows, and non-obvious architectural decisions.

Components live in multiple directories designSystem maintainability@1.0.0 Repository-wide

components, ui — keep one place for shared UI.

Remediation: Adopt shared tokens and component conventions where they help the product remain consistent.

SECTION 04

Dependency & Supply-Chain Risk

npm · Advisory check: unavailable

DIRECT PACKAGES	TRANSITIVE GRAPH	LOCKFILES INSPECTED	KNOWN ADVISORIES
62	**796**	package-lock.json	**0** matched (25 queried via GitHub Advisory Database)

Advisory lookup status: unavailable. Run network-enabled supply-chain audit in CI alongside lockfile verification.

Direct Dependency Inventory Sample (Top 24 of 62)

PACKAGE	PINNED / DECLARED VERSION	SCOPE	ECOSYSTEM / LICENSE
@apidevtools/swagger-parser	10.0.3	prod	npm
@babel/parser	7.29.8	prod	npm · MIT
@babel/traverse	7.29.8	prod	npm · MIT
@hookform/resolvers	3.10.0	prod	npm · MIT
@playwright/test	1.63.0	dev	npm · Apache-2.0
@prisma/client	5.22.0	prod	npm · Apache-2.0
@radix-ui/react-avatar	1.2.6	prod	npm · MIT
@radix-ui/react-dialog	1.1.23	prod	npm · MIT
@radix-ui/react-dropdown-menu	2.1.24	prod	npm · MIT
@radix-ui/react-label	2.1.15	prod	npm · MIT
@radix-ui/react-select	2.3.7	prod	npm · MIT
@radix-ui/react-separator	1.1.15	prod	npm · MIT
@radix-ui/react-slot	1.3.3	prod	npm · MIT
@radix-ui/react-tabs	1.1.21	prod	npm · MIT
@t3-oss/env-nextjs	0.9.2	prod	npm · MIT
@tanstack/react-query	5.105.0	prod	npm · MIT
@tanstack/react-table	8.21.3	prod	npm · MIT
@testing-library/react	14.3.1	dev	npm · MIT
@trpc/client	11.19.0	prod	npm · MIT
@trpc/react-query	11.19.0	prod	npm · MIT

PACKAGE	PINNED / DECLARED VERSION	SCOPE	ECOSYSTEM / LICENSE
@trpc/server	11.19.0	prod	npm · MIT
@types/babel__traverse	7.28.0	dev	npm · MIT
@types/dagre	0.7.54	dev	npm · MIT
@types/js-yaml	4.0.9	dev	npm